

Supplementary Material of Unsupervised Range-Nullspace Learning Prior for Multispectral Images Reconstruction

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I. APPENDIX I: ADDITIONAL EXPERIMENT RESULTS

In the supplementary material, we first present the calculation formulas for the used metrics and then provide detailed results about the experiments on different mosaic patterns, effect of different data fidelity terms, ablation study of the STV loss term, ablation study of different deep image priors, and experiments on spectral image reconstruction.

A. Metrics

This paper employs four metrics to evaluate spectral image reconstruction performance: the peak signal-to-noise ratio (PSNR), the structural similarity index measure (SSIM) [62], the Spectral Angle Mapper (SAM), and Error Relative Global Dimensionless Synthesis (ERGAS) [63].

Specifically, the PSNR is calculated as the ratio between the maximum possible intensity value of the reference b -th band image ($\text{MAX}_{\mathbf{x}_b}$) and the mean squared error (MSE) between the reconstructed and reference images. The PSNR is given by:

$$\text{PSNR} = 10 \log_{10} \left(\frac{\text{MAX}_{\mathbf{x}_b}^2}{\text{MSE}} \right). \quad (1)$$

For the b -th band image, the SSIM is calculated based on

$$\text{SSIM}(\mathbf{x}_b, \hat{\mathbf{x}}_b) = \frac{(2\mu_{\mathbf{x}_b}\mu_{\hat{\mathbf{x}}_b} + c_1)(2\sigma_{\mathbf{x}_b\hat{\mathbf{x}}_b} + c_2)}{(\mu_{\mathbf{x}_b}^2 + \mu_{\hat{\mathbf{x}}_b}^2 + c_1)(\sigma_{\mathbf{x}_b}^2 + \sigma_{\hat{\mathbf{x}}_b}^2 + c_2)}, \quad (2)$$

where $\mu_{\mathbf{x}_b}$ and $\mu_{\hat{\mathbf{x}}_b}$ denote the mean of image $\mathbf{x}_b$ and $\hat{\mathbf{x}}_b$, respectively; $\sigma_{\mathbf{x}_b}^2$ and $\sigma_{\hat{\mathbf{x}}_b}^2$ are the variances; and $\sigma_{\mathbf{x}_b\hat{\mathbf{x}}_b}$ represents the covariance between two images. The constants c_1 and c_2 are small values added to stabilize the calculation when the denominators are near zero.

For the (h, w) -th pixel, SAM is obtained by computing the angle between the reconstructed and reference spectrum

$$\text{SAM}(\mathbf{x}_{hw}, \hat{\mathbf{x}}_{hw}) = \arccos \left(\frac{\mathbf{x}_{hw} \cdot \hat{\mathbf{x}}_{hw}}{\|\mathbf{x}_{hw}\| \|\hat{\mathbf{x}}_{hw}\|} \right), \quad (3)$$

where $\mathbf{x}_{hw}$ and $\hat{\mathbf{x}}_{hw}$ are the reference and reconstructed spectral vectors, respectively. The mean of PSNR and SSIM are calculated by averaging all bands, while the mean of SAM is obtained as the average across all pixels.

Finally, ERGAS is calculated as

$$\text{ERGAS} = \frac{100}{1} \sqrt{\frac{1}{B} \sum_{b=1}^B \frac{\sum_{h,w} (\mathbf{x}_{hwb} - \hat{\mathbf{x}}_{hwb})^2}{\sum_{h,w} (\mathbf{x}_{hwb})^2}}, \quad (4)$$

where B is the total number of spectral bands, $\mathbf{x}_{hwb}$ is the pixel value at position (h, w) in the b -th spectral band of the reference image, and $\hat{\mathbf{x}}_{hwb}$ is the pixel value at position (h, w) in the b -th spectral band of the reconstructed image.

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B. Experiments on different mosaic patterns

In the supplementary material, we provide more details about the generalization ability of our UnNull method on different spectral imaging systems. As presented in the paper, we conducted experiments to evaluate the performance of the proposed UnNull algorithm on a variety of mosaic patterns, including:

- **RGB Bayer CFA:** Demosaicing experiments were performed on the classical green-blue-red-green (GBRG) RGB Bayer filter array.
- **IMEC multispectral filter arrays (3×3 and 4×4):** Beside 5×5 IMEC camera, additional experiments were conducted on IMEC 3×3 and 4×4 mosaic patterns, following the configurations described in [17].
- **MSFAs proposed by Miao *et al.* [13] and Monno *et al.* [14]:** The experiments included 4×4 multispectral filter arrays with the following configurations:

$$\begin{bmatrix} G & R & G & B \\ C & Y & C & Y \\ G & B & G & R \\ C & Y & C & Y \end{bmatrix}, \begin{bmatrix} G & R & G & Y \\ B & G & C & G \\ G & Y & G & R \\ C & G & B & G \end{bmatrix}. \quad (5)$$

The first pattern is designed by Miao *et al.* [13], and the second one is introduced in Monno *et al.* [14]. Here, G, R, B, C, and Y represent green, red, blue, cyan, and yellow bands, respectively. Finally, five bands can be obtained via these filter arrays.

As the proposed method is a generalized algorithm independent of specific mosaic patterns, we treated each pixel as an individual channel for the 2×2 RGB Bayer CFA and the 4×4 MFCAs in [13] and [14]. Therefore, we initially reconstructed 4 channels for the RGB CFA and 16 channels for the MFCAs. Finally, the spectral bands were obtained by averaging. For example, the green band in RGB images was derived as the mean of the first and fourth demosaiced images, as illustrated in Fig. S1. The main quantitative results are presented in the paper. In the supplementary material, we visualize the demosaiced images in Fig. S2. From the figure, it is evident that our proposed UnNull algorithm can achieve satisfactory results across different mosaic patterns.

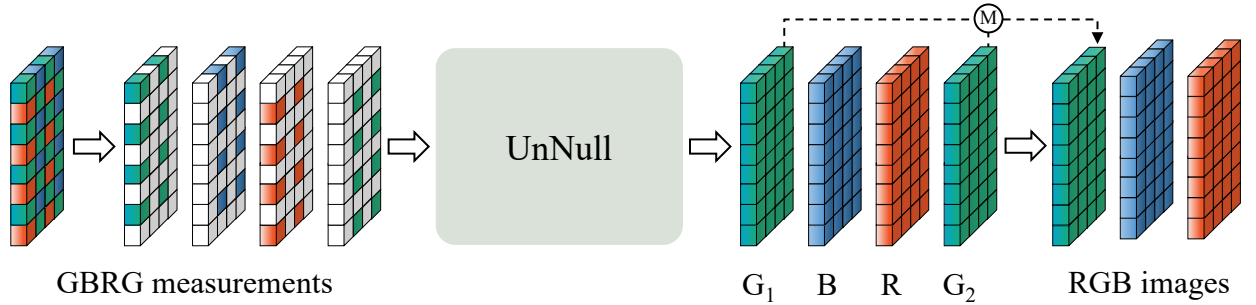


Fig. S1: The proposed UnNull algorithm for RGB images demosaicing, where ‘M’ denotes the mean of G₁ and G₂.

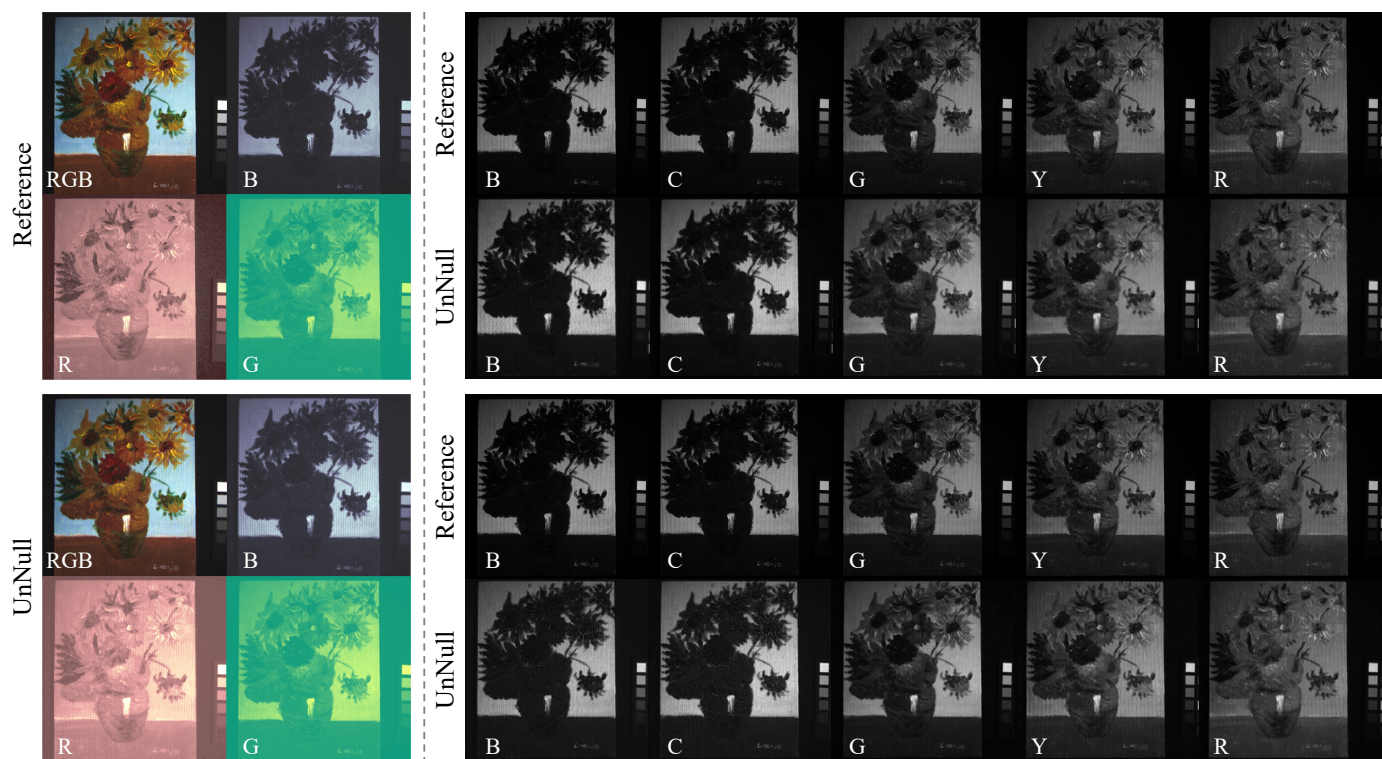


Fig. S2: Comparison of reference and UnNull demosaiced blue(B), red (R), green (G) image (left); Comparison of reference and UnNull demosaiced blue(B), cyan (C), green (G), yellow (Y), red (R) image based on the mosaic pattern proposed in [13] (right top) and mosaic pattern proposed in [14] (right bottom).

C. Effect of different data fidelity terms

In the supplementary material, we comprehensively evaluate the results of our UnNull with different data fidelity terms when measurements are corrupted by noises. Since different noise distributions ($p(e)$) require tailored data fidelity terms, we have explicitly considered the noise model via the selection of an appropriate data fidelity term $\|\cdot\|_p$ in our loss function

$$\hat{\theta} = \arg \min_{\theta} \|\mathbf{y} - \mathbf{A}\hat{\mathbf{x}}\|_p + \lambda_R \text{STV}(\hat{\mathbf{x}}), \quad (6)$$

For example, due to the foundation in the maximum likelihood estimation, the L_2 norm ($\|\cdot\|_2$) is more suitable for imaging systems affected by Gaussian noise, while the L_1/L_0 norm provides robustness to sparse noise. We then conducted experiments on 10 scenes from the CAVE dataset to evaluate the performance of our method with different data fidelity terms under various noise types. In conclusion, we evaluate the performance under Gaussian noise, sparse noise, mixed noise, and clean data in the following Table S1-S4. Results validate that:

- 1) different noise distributions have different impact on the reconstruction performance;
 - 2) utilizing the combined $L_2 + L_1$ loss can balance robustness and accuracy across diverse noise scenarios.
- Below is the empirical analysis of the effects of various noise types and fidelity terms in detail.

• Gaussian noise

We first add Gaussian noise with a level of 0.01 to the measurements $\mathbf{y}$. Gaussian noise is one of the most common types of noise encountered in imaging systems. We evaluate the proposed UnNull method with different fidelity terms. From Table S1, we can observe that our method with the L_2 norm exhibits stable performance in mitigating Gaussian noise. In addition, our method with the L_1 norm as data fidelity term also achieves competitive results. Finally, we test the combined $L_2 + L_1$ norm as loss. Results show it presents the best performance. In a nutshell, we empirically find that all the L_2 norm, L_1 norm, and the combined $L_2 + L_1$ norm can achieve satisfactory results under the Gaussian noise scenario.

Table S1: Performance comparison of the proposed method with different data fidelity terms under the condition of measurements corrupted by Gaussian noise.

Method	Scene 1	Scene 2	Scene 3	Scene 4	Scene 5	Scene 6	Scene 7	Scene 8	Scene 9	Scene 10	Average
	PSNR	PSNR	PSNR	PSNR	PSNR	PSNR	PSNR	PSNR	PSNR	PSNR	PSNR
w L_1	36.83	29.04	34.44	30.83	36.51	28.85	35.74	33.21	35.53	31.27	33.23
w L_2	36.51	29.20	34.74	31.62	36.69	28.09	36.10	33.16	35.48	31.43	33.30
w $L_2 + L_1$	37.01	29.39	34.80	31.66	36.73	28.94	36.08	33.92	35.51	31.53	33.56
	SAM	SAM	SAM	SAM	SAM	SAM	SAM	SAM	SAM	SAM	SAM
w L_1	0.149	0.206	0.185	0.245	0.375	0.188	0.046	0.193	0.384	0.200	0.217
w L_2	0.126	0.205	0.194	0.250	0.382	0.191	0.044	0.190	0.391	0.184	0.216
w $L_2 + L_1$	0.122	0.204	0.194	0.251	0.380	0.185	0.044	0.190	0.391	0.185	0.215

• Sparse noise

Then, we test the proposed UnNull method with different data fidelity terms under the measurements corrupted by sparse noise, where 0.2% of the pixels are affected (0.1% pixels are set to 1 while 0.1% pixels are set to 0). These results in Table S2 demonstrate that UnNull method with the L_1 norm as data fidelity term obviously achieves better performance than the L_2 norm. Moreover, the combination of the $L_2 + L_1$ norm as loss can also obtain superior results. In summary, we find that utilizing L_1 norm and the combined $L_2 + L_1$ norm can obtain superior performance when measurements contain sparse noise.

Table S2: Performance comparison of the proposed method with different data fidelity terms under the condition of measurements corrupted by sparse noise.

Method	Scene 1	Scene 2	Scene 3	Scene 4	Scene 5	Scene 6	Scene 7	Scene 8	Scene 9	Scene 10	Average
	PSNR	PSNR	PSNR	PSNR	PSNR	PSNR	PSNR	PSNR	PSNR	PSNR	PSNR
w L_1	37.24	29.13	34.00	29.73	34.73	28.65	34.96	31.89	36.25	29.98	32.66
w L_2	34.88	26.92	33.39	28.11	34.29	27.72	33.18	30.45	34.35	29.37	32.27
w $L_2 + L_1$	37.24	28.73	34.05	29.78	34.60	28.75	35.02	32.21	36.22	29.86	32.65
	SAM	SAM	SAM	SAM	SAM	SAM	SAM	SAM	SAM	SAM	SAM
w L_1	0.071	0.171	0.133	0.173	0.202	0.154	0.041	0.130	0.179	0.140	0.139
w L_2	0.105	0.230	0.154	0.188	0.310	0.199	0.055	0.171	0.303	0.191	0.191
w $L_2 + L_1$	0.071	0.166	0.130	0.166	0.201	0.154	0.041	0.118	0.178	0.141	0.137

- *Mixed noise (Gaussian + sparse)*

Furthermore, in real-world scenarios, measurements may contain both Gaussian and sparse noise. Mixed noise experiments are then conducted to evaluate the performance of different data fidelity terms. The noise level of Gaussian noise is 0.01 and 0.2% of the pixels are affected by sparse noise (0.1% pixels are set to 1 while 0.1% pixels are set to 0). We consider that for such cases, neither L_2 nor L_1 alone suffices. This assumption is demonstrated by the experimental results in Table S3. We can see that a combination of L_2 and L_1 can achieve superior reconstruction results when compared to the solo utilization of L_2 or L_1 norm. In conclusion, when measurements are corrupted by Gaussian and sparse mixed noise, utilizing a combination of L_2 and L_1 norm results in better results.

Table S3: Performance comparison of the proposed method with different data fidelity terms under the condition of measurements corrupted by mixed noise.

Method	Scene 1	Scene 2	Scene 3	Scene 4	Scene 5	Scene 6	Scene 7	Scene 8	Scene 9	Scene 10	Average
	PSNR	PSNR	PSNR	PSNR	PSNR	PSNR	PSNR	PSNR	PSNR	PSNR	PSNR
w L_1	32.53	27.91	31.59	29.75	33.06	27.16	32.67	30.44	33.70	29.34	30.81
w L_2	33.48	27.68	32.35	28.89	33.73	27.93	33.35	30.03	33.63	29.08	31.04
w $L_2 + L_1$	34.67	27.85	32.89	29.56	34.15	27.96	33.41	30.83	33.70	29.42	31.44
	SAM	SAM	SAM	SAM	SAM	SAM	SAM	SAM	SAM	SAM	SAM
w L_1	0.147	0.223	0.218	0.258	0.401	0.187	0.056	0.205	0.399	0.194	0.229
w L_2	0.135	0.222	0.197	0.256	0.391	0.189	0.054	0.210	0.399	0.201	0.225
w $L_2 + L_1$	0.131	0.221	0.195	0.257	0.390	0.181	0.053	0.205	0.399	0.198	0.223

- *Clean data ($e = 0$)*

Finally, we test the different data fidelity terms under the ideal noise-free scenario. This experiment serves as a baseline, allowing us to understand the inherent performance of different fidelity terms without noise interference. As shown in Table S4, the L_1 norm, the L_2 norm, and a combination of L_2 and L_1 norm are evaluated. We can see that, the L_1 norm outperformed the L_2 norm under clean conditions in most scenes. Moreover, the model with a combination of the L_2 and L_1 norm demonstrates a superior result.

Table S4: Performance comparison of the proposed method with different data fidelity terms under clean data conditions.

Method	Scene 1	Scene 2	Scene 3	Scene 4	Scene 5	Scene 6	Scene 7	Scene 8	Scene 9	Scene 10	Average
	PSNR	PSNR	PSNR	PSNR	PSNR	PSNR	PSNR	PSNR	PSNR	PSNR	PSNR
w L_1	44.56	30.07	37.94	33.57	43.81	29.16	41.48	36.95	40.29	32.09	36.99
w L_2	42.11	28.80	37.13	30.52	40.72	29.26	39.05	34.38	38.38	32.16	35.25
w $L_2 + L_1$	44.12	30.41	37.95	33.53	43.40	29.12	41.51	36.96	40.32	32.02	36.93
	SAM	SAM	SAM	SAM	SAM	SAM	SAM	SAM	SAM	SAM	SAM
w L_1	0.059	0.143	0.113	0.124	0.151	0.154	0.026	0.094	0.143	0.133	0.114
w L_2	0.066	0.185	0.113	0.153	0.168	0.180	0.034	0.123	0.161	0.178	0.136
w $L_2 + L_1$	0.048	0.143	0.113	0.125	0.151	0.154	0.026	0.095	0.143	0.132	0.113

In conclusion, considering the noise effects in real scenes and the performance in simulation experiments, we adopted the combined L_2 and L_1 norm as the data fidelity term, which can balance robustness and accuracy across diverse scenarios. In addition to the fidelity term, the used spatial-temporal total variation (STV) regularization term $STV(\hat{x})$ also can enhance the robustness of our method. While the data fidelity term handles discrepancies caused by noise in $\mathbf{y}$, the regularization term enforces smoothness and sparsity in the reconstruction $\hat{x}$.

D. Ablation study of the STV loss term

In the supplementary material, the detailed experiment comparing the performance of our method with and without the STV loss term across 10 scenes from the CAVE dataset using various metrics, including PSNR, SSIM, SAM, and ERGAS, are presented. The experimental results are summarized in the Table S5. From these results, it is evident that the STV regularization term significantly enhances the spatial and spectral quality of the reconstructed images, as demonstrated by consistent improvements across all evaluated metrics.

Table S5: Quantitative comparison of PSNR, SSIM, SAM (mean $\pm$ standard deviation), and ERGAS results for the proposed method with and without the STV loss term across 10 scenes.

Method	Scene 1	Scene 2	Scene 3	Scene 4	Scene 5	Scene 6	Scene 7	Scene 8	Scene 9	Scene 10	Average
	PSNR	PSNR	PSNR	PSNR	PSNR	PSNR	PSNR	PSNR	PSNR	PSNR	PSNR
Ours	44.12$\pm$1.00	30.41$\pm$1.48	37.95$\pm$1.32	33.53$\pm$1.42	43.40$\pm$2.75	29.12$\pm$2.02	41.51$\pm$0.93	36.96$\pm$1.86	40.32$\pm$2.50	32.02$\pm$1.08	36.93
w/o STV	43.07 $\pm$ 1.40	29.41 $\pm$ 1.87	36.65 $\pm$ 1.62	30.92 $\pm$ 3.44	42.81 $\pm$ 3.36	25.30 $\pm$ 2.52	40.89 $\pm$ 1.84	34.98 $\pm$ 2.18	38.67 $\pm$ 2.97	29.80 $\pm$ 3.79	35.25
	SSIM	SSIM	SSIM	SSIM	SSIM	SSIM	SSIM	SSIM	SSIM	SSIM	SSIM
Ours	0.999$\pm$0.00	0.989$\pm$0.00	0.995$\pm$0.00	0.992$\pm$0.02	0.994$\pm$0.00	0.984$\pm$0.02	0.997$\pm$0.00	0.994$\pm$0.00	0.994$\pm$0.03	0.975$\pm$0.01	0.991
w/o STV	0.999 $\pm$ 0.00	0.982 $\pm$ 0.01	0.993 $\pm$ 0.00	0.983 $\pm$ 0.01	0.991 $\pm$ 0.00	0.955 $\pm$ 0.01	0.995 $\pm$ 0.00	0.992 $\pm$ 0.00	0.991 $\pm$ 0.02	0.941 $\pm$ 0.02	0.982
	SAM	SAM	SAM	SAM	SAM	SAM	SAM	SAM	SAM	SAM	SAM
Ours	0.048$\pm$0.11	0.143$\pm$0.14	0.113$\pm$0.12	0.125$\pm$0.14	0.151$\pm$0.17	0.154$\pm$0.10	0.026$\pm$0.08	0.095$\pm$0.13	0.143$\pm$0.15	0.132$\pm$0.12	0.113
w/o STV	0.051 $\pm$ 0.12	0.157 $\pm$ 0.15	0.120 $\pm$ 0.12	0.135 $\pm$ 0.14	0.153 $\pm$ 0.17	0.191 $\pm$ 0.13	0.028 $\pm$ 0.09	0.101 $\pm$ 0.13	0.152 $\pm$ 0.15	0.174 $\pm$ 0.13	0.126
	ERGAS	ERGAS	ERGAS	ERGAS	ERGAS	ERGAS	ERGAS	ERGAS	ERGAS	ERGAS	ERGAS
Ours	03.72	26.74	18.83	14.01	14.79	21.77	02.89	12.17	15.59	17.84	14.83
w/o STV	04.41	29.12	19.67	19.05	15.64	40.66	03.23	15.37	16.72	27.04	19.09

E. Ablation study of different deep image priors

In the supplementary material, we provided the detailed results for comparing our proposed method with the two suggested variations: 1) $\hat{x} = f_\theta(Gy)$, 2) $\hat{x} = Gy + f_\theta(Gy)$. The experiments were performed on 10 scenes from the CAVE dataset, with results evaluated using PSNR, SSIM, SAM, and ERGAS metrics. Detailed results are presented in Table S6. From the results, it can be observed that our proposed method consistently outperforms the two variations in terms of all evaluation metrics across all scenes.

Table S6: Quantitative comparison of PSNR, SSIM, SAM (mean $\pm$ standard deviation), and ERGAS results across 10 scenes from the CAVE dataset. Results of our proposed method and the two variations are shown.

Method	Scene 1	Scene 2	Scene 3	Scene 4	Scene 5	Scene 6	Scene 7	Scene 8	Scene 9	Scene 10	Average
	PSNR	PSNR	PSNR	PSNR	PSNR	PSNR	PSNR	PSNR	PSNR	PSNR	PSNR
Ours	44.12$\pm$1.00	30.41$\pm$1.48	37.95$\pm$1.32	33.53$\pm$1.42	43.40$\pm$2.75	29.12$\pm$2.02	41.51$\pm$0.93	36.96$\pm$1.86	40.32$\pm$2.50	32.02$\pm$1.08	36.93
$Gy + f_\theta(Gy)$	38.15 $\pm$ 1.69	26.62 $\pm$ 2.10	35.26 $\pm$ 1.37	29.60 $\pm$ 3.15	37.23 $\pm$ 3.71	26.15 $\pm$ 2.17	36.27 $\pm$ 2.14	31.49 $\pm$ 2.25	36.12 $\pm$ 2.42	29.60 $\pm$ 3.42	32.65
$f_\theta(Gy)$	41.48 $\pm$ 1.41	30.05 $\pm$ 1.15	35.45 $\pm$ 1.75	32.99 $\pm$ 1.86	40.31 $\pm$ 2.42	28.16 $\pm$ 2.04	38.48 $\pm$ 1.26	35.97 $\pm$ 1.45	39.53 $\pm$ 2.17	30.80 $\pm$ 3.11	35.32
	SSIM	SSIM	SSIM	SSIM	SSIM	SSIM	SSIM	SSIM	SSIM	SSIM	SSIM
Ours	0.999$\pm$0.00	0.989$\pm$0.00	0.995$\pm$0.00	0.992$\pm$0.02	0.994$\pm$0.00	0.984$\pm$0.02	0.997$\pm$0.00	0.994$\pm$0.00	0.994$\pm$0.03	0.975$\pm$0.01	0.991
$Gy + f_\theta(Gy)$	0.996 $\pm$ 0.01	0.969 $\pm$ 0.02	0.990 $\pm$ 0.00	0.975 $\pm$ 0.01	0.955 $\pm$ 0.02	0.965 $\pm$ 0.01	0.988 $\pm$ 0.00	0.981 $\pm$ 0.01	0.987 $\pm$ 0.01	0.939 $\pm$ 0.02	0.974
$f_\theta(Gy)$	0.998 $\pm$ 0.03	0.983 $\pm$ 0.02	0.985 $\pm$ 0.01	0.983 $\pm$ 0.03	0.990 $\pm$ 0.01	0.984$\pm$0.01	0.994 $\pm$ 0.00	0.994$\pm$0.00	0.991 $\pm$ 0.03	0.970 $\pm$ 0.01	0.987
	SAM	SAM	SAM	SAM	SAM	SAM	SAM	SAM	SAM	SAM	SAM
Ours	0.048$\pm$0.11	0.143$\pm$0.14	0.113$\pm$0.12	0.125$\pm$0.14	0.151$\pm$0.17	0.154$\pm$0.10	0.026$\pm$0.08	0.095 $\pm$ 0.13	0.143$\pm$0.15	0.132$\pm$0.12	0.113
$Gy + f_\theta(Gy)$	0.069 $\pm$ 0.14	0.216 $\pm$ 0.19	0.122 $\pm$ 0.13	0.161 $\pm$ 0.16	0.176 $\pm$ 0.19	0.184 $\pm$ 0.13	0.039 $\pm$ 0.10	0.139 $\pm$ 0.18	0.184 $\pm$ 0.18	0.178 $\pm$ 0.14	0.147
$f_\theta(Gy)$	0.059 $\pm$ 0.11	0.146 $\pm$ 0.13	0.114 $\pm$ 0.11	0.127 $\pm$ 0.13	0.159 $\pm$ 0.16	0.161 $\pm$ 0.10	0.032 $\pm$ 0.08	0.087$\pm$0.11	0.145 $\pm$ 0.15	0.137 $\pm$ 0.11	0.117
	ERGAS	ERGAS	ERGAS	ERGAS	ERGAS	ERGAS	ERGAS	ERGAS	ERGAS	ERGAS	ERGAS
Ours	03.72	26.74	18.83	14.01	14.79	21.77	02.89	12.17	15.59	17.84	14.83
$Gy + f_\theta(Gy)$	08.66	47.07	26.26	21.99	33.86	36.55	05.49	22.83	25.39	27.44	25.55
$f_\theta(Gy)$	06.02	27.25	25.86	18.90	16.42	22.33	04.34	13.55	16.10	23.56	17.43

F. Visualization of spectral image reconstruction

In the supplementary material, we visualized the spectral images reconstructed by our UnNull method in Fig. S3. By comparing with the reference images, we can observe that our method can generate accurate results, which demonstrates UnNull's ability in the spectral image reconstruction task.

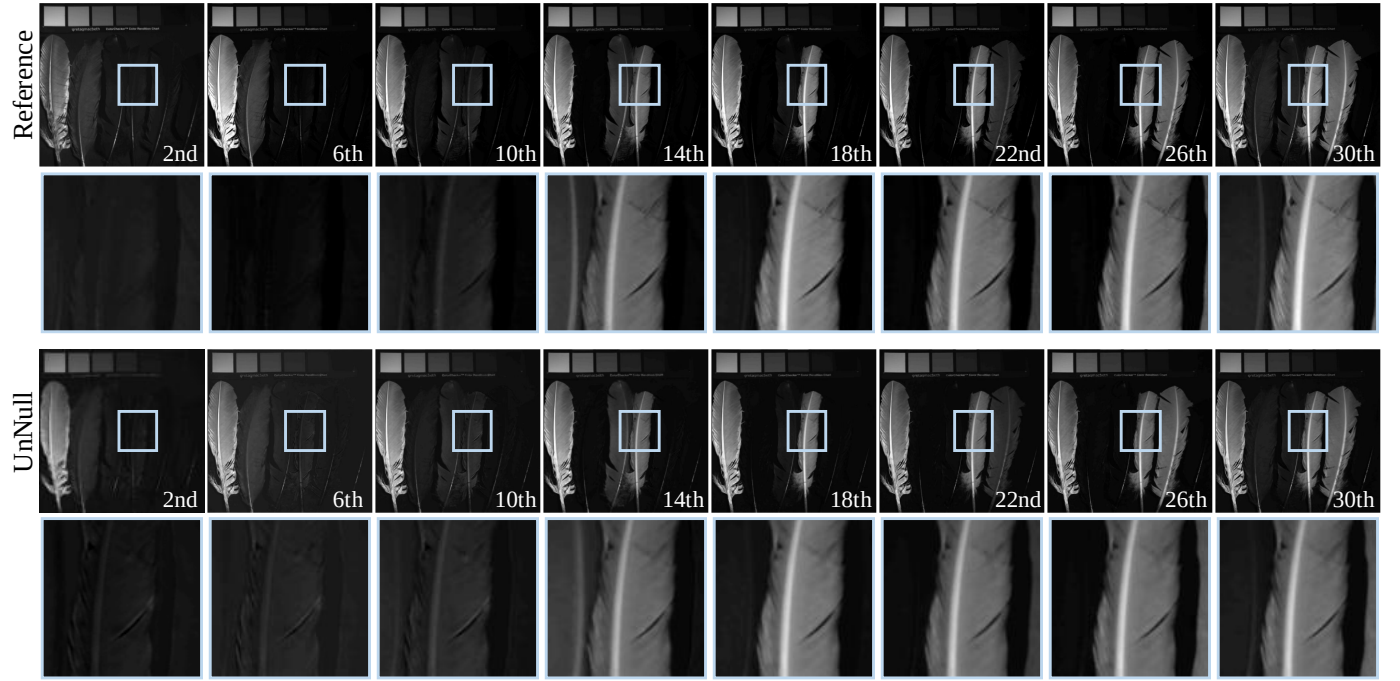


Fig. S3: Comparison of spectral images reconstructed using the proposed UnNull method and the reference. Eight out of 31 bands were selected for visualization.



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